changes regarding roles (Chapter12/12-end/polar-deployment/docker/keycloak/full-realm-config.json). I recommend updating your `polar-keycloak` container definition in Docker Compose to use this new JSON file.

Before moving on to the next section, stop any running containers (`docker-compose down`).

CONFIGURING ACCESS TO USER ROLES IN SPRING SECURITY

Keycloak is now configured to return the authenticated user roles in a `roles` claim included in both the ID Token and Access Token. However, the `roles` claim will only be returned if the OAuth2 Client (Edge Service) asks for the `roles` scope.

In the Edge Service project, open the application.yml file, and update the Client Registration configuration to include the `roles` scope.

Listing 12.3 Assigning the `roles` scope to Edge Service

```
spring:
  security:
    oauth2:
      client:
        registration:
          keycloak:
            client-id: edge-service
            client-secret: polar-keycloak-secret
            scope: openid,roles                    ❶
        provider:
          keycloak:
            issuer-uri: http://localhost:8080/realms/PolarBookshop
```

❶ Adds “roles” to the list of scopes so that Edge Service can get access to the user roles

Next you’ll see how to extract the roles for the currently authenticated user from the ID Token.